

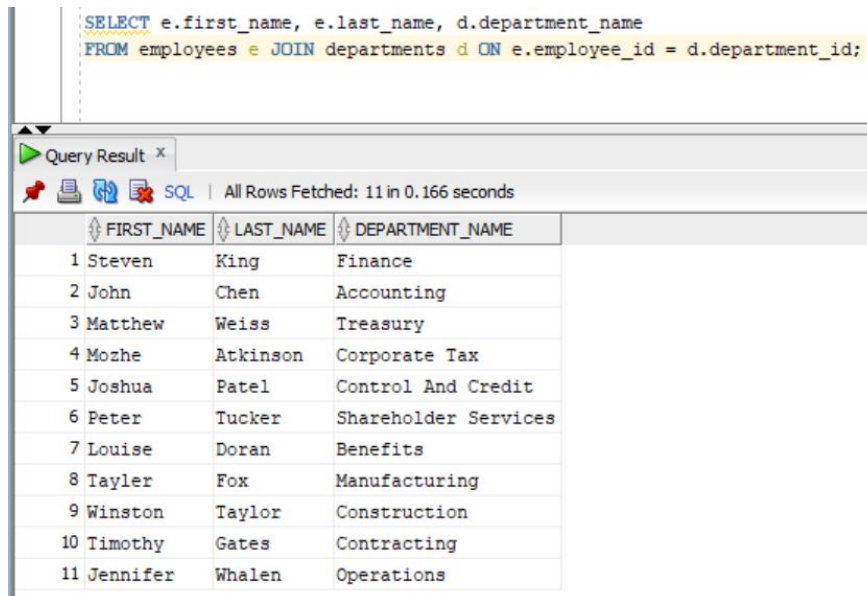
Oracle SQL Code Review & Query Logic Validation

Query Review 1

Original Query

```
SELECT e.first_name, e.last_name, d.department_name  
FROM employees e JOIN departments d ON e.employee_id = d.department_id;
```

Output: Query executed successfully and returned rows.



The screenshot shows an Oracle SQL Developer window. At the top, the SQL editor contains the query: `SELECT e.first_name, e.last_name, d.department_name FROM employees e JOIN departments d ON e.employee_id = d.department_id;`. Below the editor, the 'Query Result' tab is active, displaying the results of the query. The status bar indicates 'All Rows Fetched: 11 in 0.166 seconds'. The results are shown in a table with three columns: FIRST_NAME, LAST_NAME, and DEPARTMENT_NAME. The table contains 11 rows of data, each with a row number in the first column.

	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	Steven	King	Finance
2	John	Chen	Accounting
3	Matthew	Weiss	Treasury
4	Mozhe	Atkinson	Corporate Tax
5	Joshua	Patel	Control And Credit
6	Peter	Tucker	Shareholder Services
7	Louise	Doran	Benefits
8	Tayler	Fox	Manufacturing
9	Winston	Taylor	Construction
10	Timothy	Gates	Contracting
11	Jennifer	Whalen	Operations

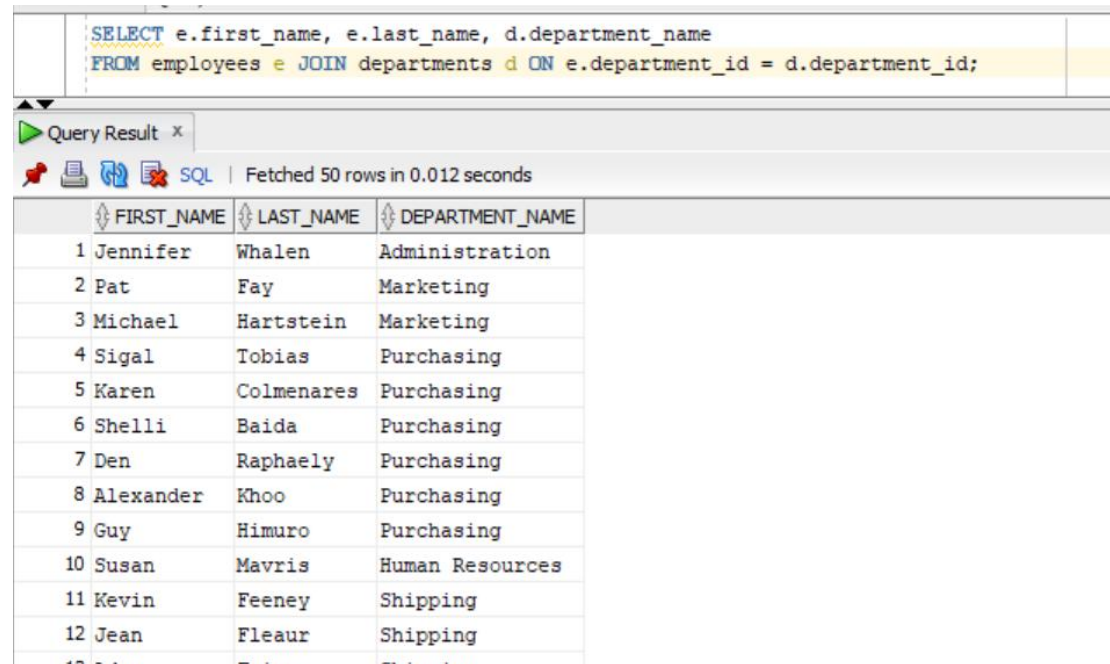
Business Logic Analysis

- The query result is not logically correct.
- Employee ID should not be matched with Department ID.
- These columns represent different entities and are unrelated.
- The returned rows are coincidental matches rather than valid business relationships.

Correct Query

```
SELECT e.first_name, e.last_name, d.department_name
```

```
FROM employees e JOIN departments d ON e.department_id = d.department_id;
```



The screenshot shows a SQL query execution window. The query is: `SELECT e.first_name, e.last_name, d.department_name FROM employees e JOIN departments d ON e.department_id = d.department_id;`. The results are displayed in a table with 50 rows. The first 13 rows are visible, showing employee names and their department names.

	FIRST_NAME	LAST_NAME	DEPARTMENT_NAME
1	Jennifer	Whalen	Administration
2	Pat	Fay	Marketing
3	Michael	Hartstein	Marketing
4	Sigal	Tobias	Purchasing
5	Karen	Colmenares	Purchasing
6	Shelli	Baida	Purchasing
7	Den	Raphaely	Purchasing
8	Alexander	Khoo	Purchasing
9	Guy	Himuro	Purchasing
10	Susan	Mavris	Human Resources
11	Kevin	Feeney	Shipping
12	Jean	Fleaur	Shipping
13	...	...	...

Reasoning

- Employees belong to departments through the DEPARTMENT_ID foreign key.
- The corrected query uses the proper relationship between EMPLOYEES and DEPARTMENTS.
- This returns accurate employee and department information.

Conclusion

- SQL Syntax: Valid
- Business Logic: Incorrect
- Improvement Required: Yes

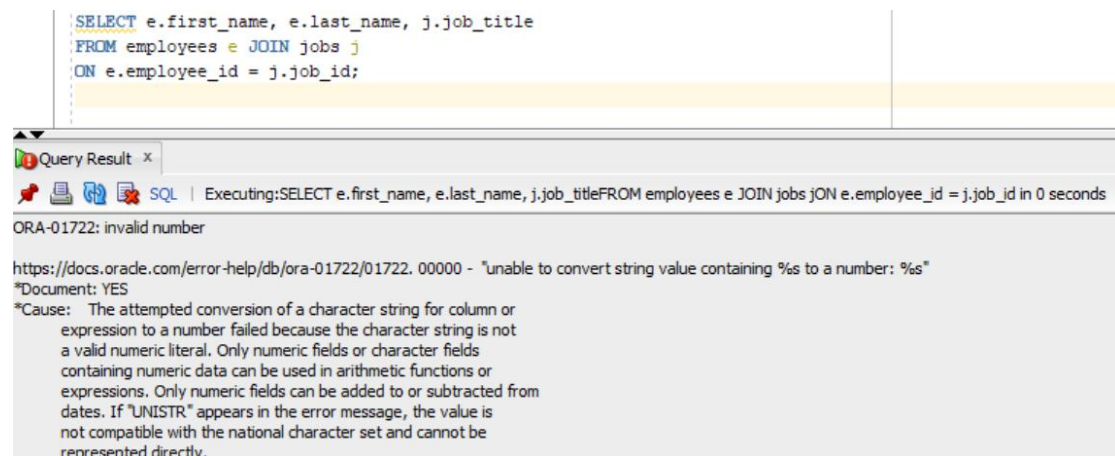
Query Review 2

Original Query

```
SELECT e.first_name, e.last_name, j.job_title FROM employees e JOIN jobs j  
ON e.employee_id = j.job_id;
```

Output

ORA-01722: invalid number



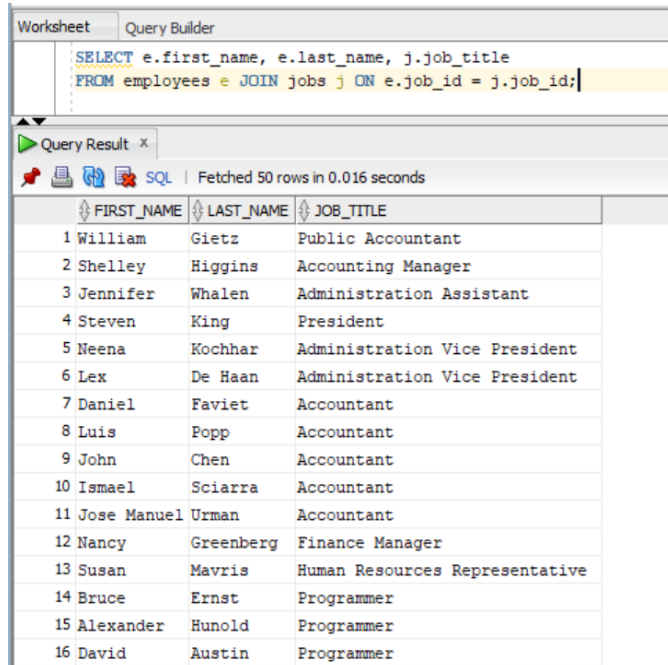
Business Logic Analysis

- Employee ID is a numeric column.
- Job ID is a character column.
- The query attempts to compare unrelated columns with different data types.
- Therefore Oracle raises an invalid number error.

Correct Query

```
SELECT e.first_name, e.last_name, j.job_title
```

```
FROM employees e JOIN jobs j ON e.job_id = j.job_id;
```



The screenshot shows a database query tool interface. The top tab is 'Worksheet' and the bottom tab is 'Query Builder'. The SQL query entered is: `SELECT e.first_name, e.last_name, j.job_title FROM employees e JOIN jobs j ON e.job_id = j.job_id;`. Below the query, the 'Query Result' tab is active, showing 'Fetched 50 rows in 0.016 seconds'. The results are displayed in a table with columns: FIRST_NAME, LAST_NAME, and JOB_TITLE. The table contains 16 rows of data.

	FIRST_NAME	LAST_NAME	JOB_TITLE
1	William	Gietz	Public Accountant
2	Shelley	Higgins	Accounting Manager
3	Jennifer	Whalen	Administration Assistant
4	Steven	King	President
5	Neena	Kochhar	Administration Vice President
6	Lex	De Haan	Administration Vice President
7	Daniel	Faviet	Accountant
8	Luis	Popp	Accountant
9	John	Chen	Accountant
10	Ismael	Sciarra	Accountant
11	Jose Manuel	Urman	Accountant
12	Nancy	Greenberg	Finance Manager
13	Susan	Mavris	Human Resources Representative
14	Bruce	Ernst	Programmer
15	Alexander	Hunold	Programmer
16	David	Austin	Programmer

Reasoning

- Employees are linked to jobs through JOB_ID.
- The corrected query uses the proper foreign key relationship.
- This returns the correct job title for each employee.

Conclusion

- SQL Syntax: Valid
- Business Logic: Incorrect
- Improvement Required: Yes

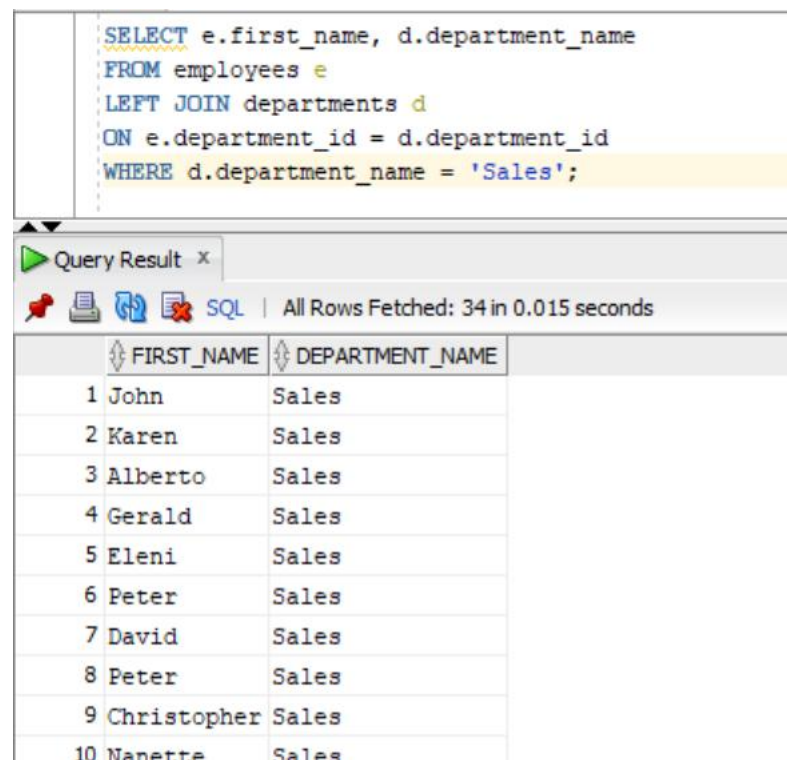
Query Review 3

Original Query

```
SELECT e.first_name, d.department_name FROM employees e LEFT JOIN  
departments d ON e.department_id = d.department_id WHERE  
d.department_name = 'Sales';
```

Output

Query executed successfully and returned employees working in the Sales department.



The screenshot shows a SQL query editor with the following query:

```
SELECT e.first_name, d.department_name  
FROM employees e  
LEFT JOIN departments d  
ON e.department_id = d.department_id  
WHERE d.department_name = 'Sales';
```

Below the query editor, the 'Query Result' window is displayed, showing the results of the query. The window indicates that all rows were fetched in 0.015 seconds. The results are as follows:

	FIRST_NAME	DEPARTMENT_NAME
1	John	Sales
2	Karen	Sales
3	Alberto	Sales
4	Gerald	Sales
5	Eleni	Sales
6	Peter	Sales
7	David	Sales
8	Peter	Sales
9	Christopher	Sales
10	Nanette	Sales

Analysis

- The query is syntactically correct and produces the expected result.
- However, the LEFT JOIN is unnecessary because the WHERE condition filters only rows where department_name = 'Sales'.
- This makes the LEFT JOIN behave like an INNER JOIN.

-Suggested Improvement

```
SELECT e.first_name, d.department_name FROM employees e JOIN  
departments d ON e.department_id = d.department_id WHERE  
d.department_name = 'Sales';
```

<pre> SELECT e.first_name, d.department_name FROM employees e JOIN departments d ON e.department_id = d.department_id WHERE d.department_name = 'Sales'; </pre>		
 Query Result x  All Rows Fetched: 34 in 0.013 seconds		
	FIRST_NAME	DEPARTMENT_NAME
1	John	Sales
2	Karen	Sales
3	Alberto	Sales
4	Gerald	Sales
5	Eleni	Sales
6	Peter	Sales
7	David	Sales
8	Peter	Sales
9	Christopher	Sales
10	Nanette	Sales
11	Oliver	Sales
12	Janette	Sales
13	Patrick	Sales

Reasoning

- INNER JOIN is more appropriate because only matching departments
- are required in the result.
- It makes the query easier to understand and more efficient.

Subquery Type: No subquery used.

Execution Logic

1. EMPLOYEES is joined with DEPARTMENTS using DEPARTMENT_ID.
2. Rows are filtered to keep only the Sales department.
3. Matching employee names and department names are returned.

Conclusion

- SQL Syntax: Correct
- Business Logic: Correct
- Optimization Suggested: Use INNER JOIN instead of LEFT JOIN.

Query Review 4

Original Query

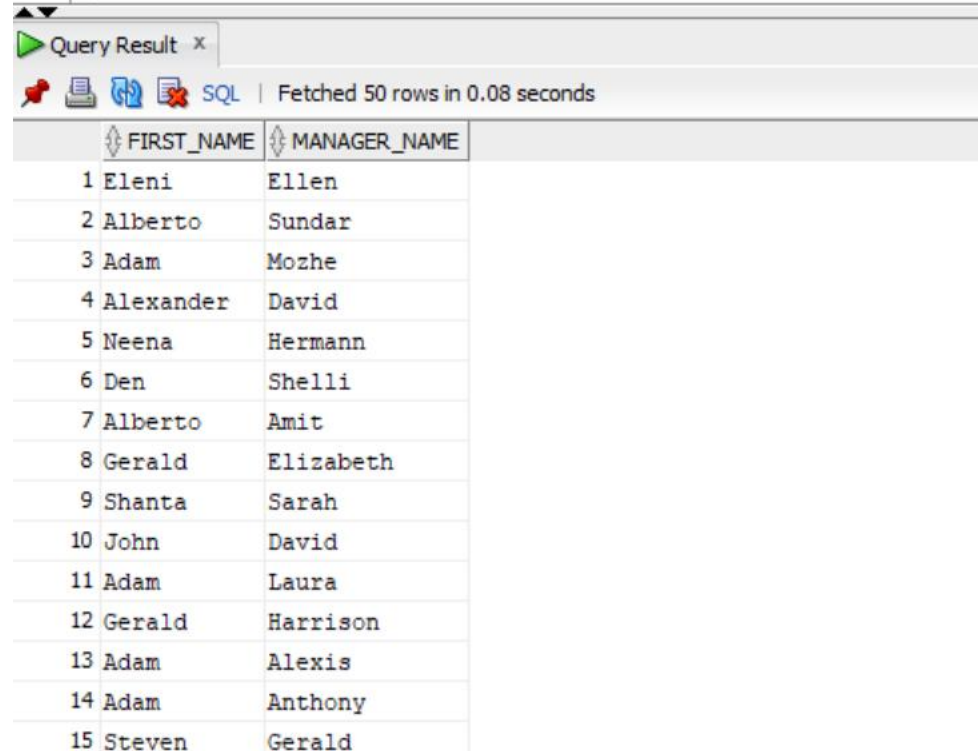
```
SELECT e.first_name, m.first_name AS manager_name
```

```
FROM employees e JOIN employees m ON e.employee_id = m.manager_id;
```

Output

Query executed successfully and returned rows.

```
SELECT e.first_name, m.first_name AS manager_name
FROM employees e
JOIN employees m
ON e.employee_id = m.manager_id;
```



The screenshot shows a SQL query result window titled "Query Result x". It displays the results of the query: `SELECT e.first_name, m.first_name AS manager_name FROM employees e JOIN employees m ON e.employee_id = m.manager_id;`. The window indicates that 50 rows were fetched in 0.08 seconds. The results are shown in a table with two columns: `FIRST_NAME` and `MANAGER_NAME`. The table contains 15 rows of data, each with an employee's first name and their manager's first name.

	FIRST_NAME	MANAGER_NAME
1	Eleni	Ellen
2	Alberto	Sundar
3	Adam	Mozhe
4	Alexander	David
5	Neena	Hermann
6	Den	Shelli
7	Alberto	Amit
8	Gerald	Elizabeth
9	Shanta	Sarah
10	John	David
11	Adam	Laura
12	Gerald	Harrison
13	Adam	Alexis
14	Adam	Anthony
15	Steven	Gerald

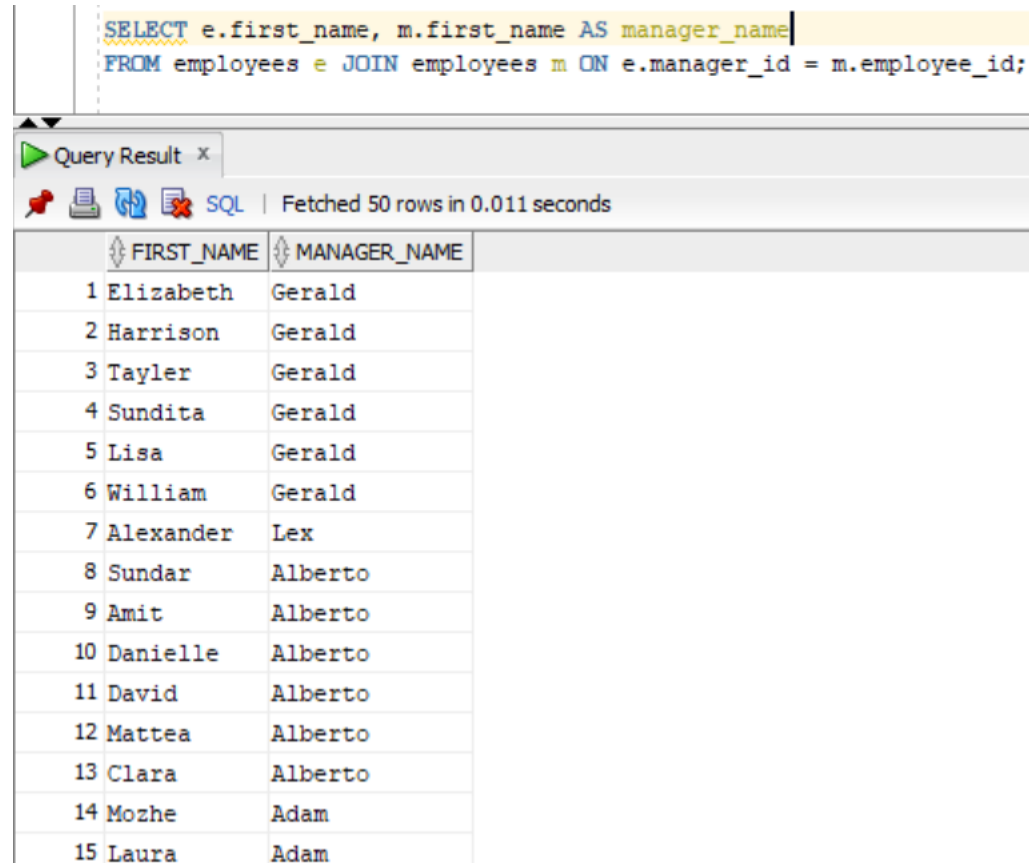
Business Logic Analysis

- The query is syntactically correct, but the self-join condition is incorrect.
- It matches `employee_id` with `manager_id` in the wrong direction.
- This returns employees who manage others, instead of returning each employee with their direct manager.

Correct Query

SELECT e.first_name, m.first_name AS manager_name

FROM employees e JOIN employees m ON e.manager_id = m.employee_id;



The screenshot shows a SQL query execution window. The query is: `SELECT e.first_name, m.first_name AS manager_name FROM employees e JOIN employees m ON e.manager_id = m.employee_id;`. The results are displayed in a table with 15 rows. The first column is labeled 'FIRST_NAME' and the second column is labeled 'MANAGER_NAME'. The data shows that employees 1 through 6 are managed by Gerald, employees 7 through 13 are managed by Alberto, and employees 14 and 15 are managed by Adam.

	FIRST_NAME	MANAGER_NAME
1	Elizabeth	Gerald
2	Harrison	Gerald
3	Tayler	Gerald
4	Sundita	Gerald
5	Lisa	Gerald
6	William	Gerald
7	Alexander	Lex
8	Sundar	Alberto
9	Amit	Alberto
10	Danielle	Alberto
11	David	Alberto
12	Mattea	Alberto
13	Clara	Alberto
14	Mozhe	Adam
15	Laura	Adam

Reasoning

- In the EMPLOYEES table, MANAGER_ID stores the employee ID of the manager.
- Therefore, employee manager_id should match the manager employee_id.

Conclusion

- SQL Syntax: Correct
- Business Logic: Incorrect
- Improvement Required: Yes